Power Series Solutions to Nonlinear Ordinary Differential Equations
MATH 689 Homework 3

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1. From problems 2.2e (p. 45) and 3.12 (p. 81):

$$f = \frac{e^x}{1+x} = \sum_{n=0}^{\infty} c_n x^n, \quad |x| < R \tag{1}$$

(a) Determine an expression for all c_n's using Cauchy's product rule and the knowledge that

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n \quad \text{and} \quad \frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n. \tag{2}$$

Let $a_n = \frac{1}{n!}$ and $b_n = (-1)^n$, then from the Cauchy's product we have c_n as

$$c_n = \sum_{0 \le j \le n} \frac{(-1)^{n-j}}{j!} = (-1)^n \sum_{0 \le j \le n} \frac{(-1)^j}{j!} \tag{3}$$

(b) Determine an alternate recursive expression for all c_n's using the identity for dividing two infinite series, noting that the denominator of f may be thought of as an infinite series made up of mostly zero-valued coefficients.

The identity for dividing two infinite series of recursion

$$c_n = \frac{1}{b_0} \left[a_n - \sum_{1 \le j \le n} b_j c_{n-j} \right] \tag{4}$$

Applying this on (1) with $b_0 = b_1 = 1, b_{n>1} = 0$ yields

$$c_n = \frac{1}{n!} - \left(c_{n-1} + \sum_{2 \le j \le n} b_j c_{n-j} \right) = \frac{1}{n!} - c_{n-1}, \quad c_0 = 1, \quad n \ge 1 \tag{5}$$

(c) Check your expressions by confirming that c_0 through c_4 obtained from (a) match the values of c_0 through c_4 obtained from (b).

Using the recursion obtained in (3) we get

$$\{c_n\}_{n=0}^{4} = \{c_0, c_1, c_2, c_3, c_4\} \tag{6}$$

$$\begin{aligned} &c_0 = (-1)^0 \sum_{0 \le j \le 0} \frac{(-1)^j}{j!} = 1, \quad c_1 = (-1)^1 \sum_{0 \le j \le 1} \frac{(-1)^j}{j!} = 0 \\ &c_2 = (-1)^2 \sum_{0 \le j \le 2} \frac{(-1)^j}{j!} = \frac{1}{2}, \quad c_3 = (-1)^3 \sum_{0 \le j \le 3} \frac{(-1)^j}{j!} = -\frac{1}{3} \\ &c_4 = (-1)^4 \sum_{0 \le j \le 4} \frac{(-1)^j}{j!} = \frac{3}{8} \end{aligned} \tag{7}$$

Then using the recursion in (5)

$$\begin{aligned} c_0 = 1, \quad c_1 = 1 - 1 = 0, \quad c_2 = \frac{1}{2} - 0 = \frac{1}{2} \\ c_3 = \frac{1}{6} - \frac{3}{6} = -\frac{1}{3}, \quad c_4 = \frac{1}{24} + \frac{1}{3} = \frac{3}{8} \end{aligned} \tag{8}$$

We see that the coefficients $\{c_n\}_{n=0}^{4}$ from both parts agree.

(d) Using either expression ((a) or (b)), plot the series and compare truncations of it with the actual function (you may SeriesDivisionExample.m as a template). We plot the partial sums with N terms for $N : N \in \mathbb{N}, \quad 1 \leq N \leq 100, \quad N \equiv 0 \pmod{10}$.

$$f_{S,N}(x) = 1 + \sum_{n=1}^{N} \left(\frac{1}{n!} - c_{n-1} \right) x^n, \quad |x| < \limsup \left| \frac{c_n}{c_{n+1}} \right| = R \tag{9}$$

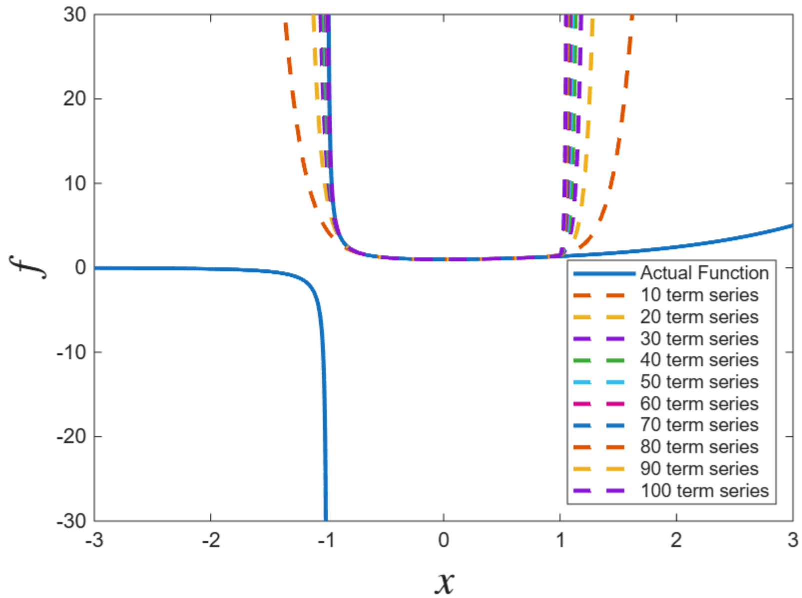

Figure 1: Truncated Solution (5) vs Actual Function (1)

(e) Estimate R using the ratio test, plotting $\left|\frac{c_n}{c_{n+1}}\right|$ versus n and seeking the limiting value as $n \to \infty$. You may add the code snippet below to the code you used in (d).

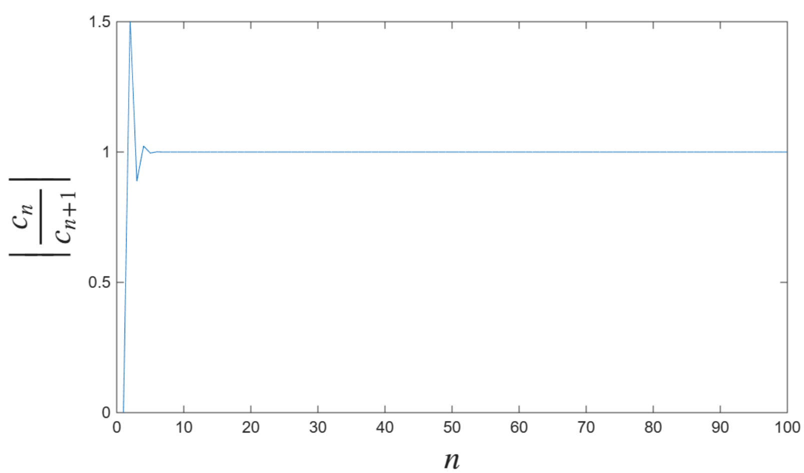

Figure 2: Ratio Test to approximate R: $\limsup |c_n/c_{n+1}|$

(f) Estimate R using the root test, plotting $|c_n|^{-1/n}$ versus n and seeking the limiting value as $n \to \infty$. You may add the code snippet below to replace that used in (e).

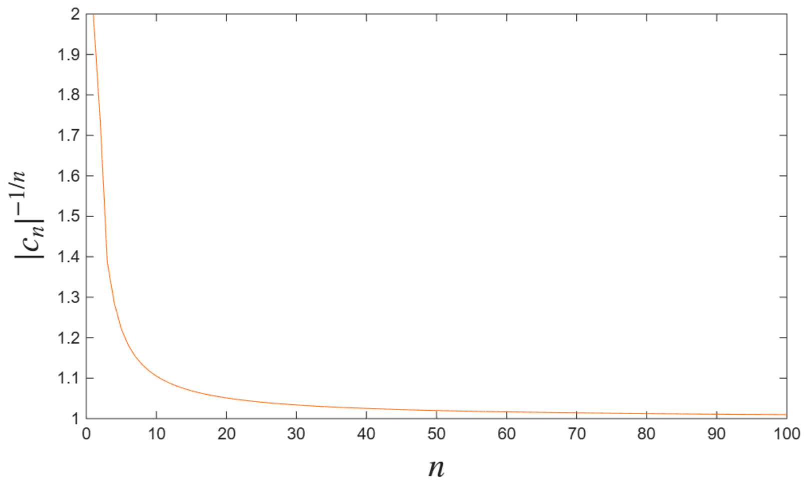

Figure 3: Root Test: $\limsup |c_n|^{-1/n}$

(g) In (e) and (f), how did you decide how many terms (N) to take in the series for each test?

This will be apparent in the MATLAB code, to be explicit, the N chosen here is the same N used in the maximum number of terms in the truncated solution, this was what we conclude about the convergences up to N is reflected in the truncated power series solution. We can see that with the same N, the transient effects fades away and we see that both root test and ratio test converge to the same value $R = 1$, thus further confirming why we have a singularity at $x = -1$. Our truncated series solution (9) is centered around $x = 0$, with radius $R \approx 1$

2. HW1 Problem 4 revisited (The Flierl-Petviashvili model [1]) [from problem 3.7, p. 80]

$$f'' + \frac{1}{x}f' - f - f^2 = 0, \quad f(0) = -2.392\ldots, \quad f'(0) = 0, \quad x \in [0, 10] \tag{10}$$

(a) Find the power series solution of the ODE. That is, assume $f = \sum_{n=0}^{\infty} a_n x^n$ and determine a recurrence relation for the a_n's. Show that the answer is

$$f = \sum_{n=0}^{\infty} a_n x^n, \quad a_{n+2} = \frac{a_n + \sum_{j=0}^{n} a_j a_{n-j}}{(n+2)^2}, \quad a_1 = 0, \quad a_0 = -2.392. \tag{11}$$

(Hint: Evaluate the x^{-1} terms on each side separately before evaluating the x^n terms for $n \geq 0$.)

Plugging f, f' and f'' yields the following

$$\sum_{n\geq 0} \left[a_{n+2}(n+2)(n+1) + a_{n+2}(n+2) - a_n - \sum_{0\leq j\leq n} a_j a_{n-j}\right]x^n + \frac{a_1}{x} = 0 \tag{12}$$

Since (11) is 0 on the LHS so we know $a_1 = 0$, and $a_0 = -2.392$ from the initial conditions, then solving for a_{n+2} gives the same as proposed above

$$a_{n+2} = \frac{a_n + \sum_{0\leq j\leq n} a_j a_{n-j}}{(n+2)^2}, \quad a_1 = 0, \quad a_0 = -2.392. \tag{13}$$

(b) Incorporate the power series solution into the code you wrote for HW1Problem4. Example lines (for you to modify) using Cauchy's product rule are provided in lines 50-56 of HW1Problem1withSeries.m.

(c) Plot successive truncations of the series on the same graph as the numerical solution. Estimate the radius of convergence, R, based on this plot; in other words, for what x value (and beyond), does the series not agree with the numerical solution, no matter how many terms you use?

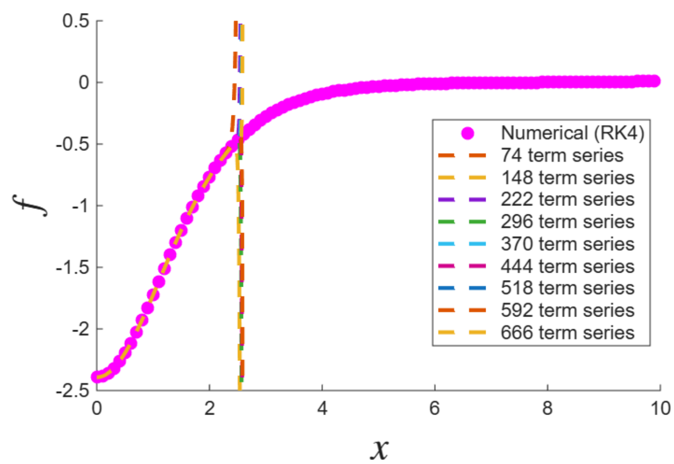

Figure 4: Truncated Solution ((11) vs RK4 Solution to (10)

We plot $f_{S,N}$ for $N : N \in \mathbb{N}, \quad 1 \leq N \leq 674, \quad N \equiv 0 \pmod{74}$. We acknowledge that for even power terms we have that the truncated sum diverges to negative infinity at approximately R, and for odd power terms the truncated sum diverges to positive infinity at approximately R. From MATLAB we can approximate that this R is ≈ 2.5843

(d) Use the root test to estimate R, plotting $|a_n|^{-1/n}$ versus n and seeking the limiting value as $n \to \infty$. You can skip over the zero-valued odd coefficients using the code snippet:

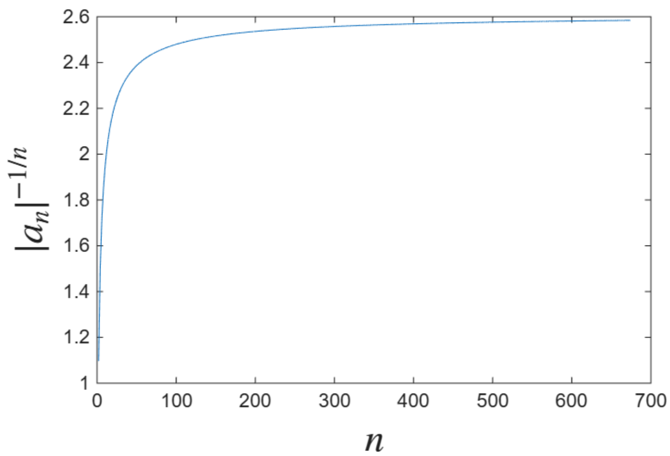

Figure 5: Root Test: $\limsup |a_n|^{-1/n}$

(e) In (d), how did you decide how many terms (N) to take in the series for each test?

For part (d) I choose the same $N = 674$ as I did for the truncated series solution as it would mean what we say about the root test holds for the series up to the same N. Also this N illustrates the convergence of the radius of $R \approx 2.58409$.

3. MATH 689 Problem (HW2 Problem 4 revisited), from problem 2.15 on p. 47:

$$f' + f = \frac{1}{1+x}, \quad f(0) = 1, \quad x \in [0, 2] \tag{14}$$

(a) Estimate the radius of convergence, R, of the series solution, $f = \sum_{n=0}^{\infty} a_n x^n$ using

i. the ratio test, plotting $\left|\frac{a_n}{a_{n+1}}\right|$ versus n and seeking the limiting value as $n \to \infty$.

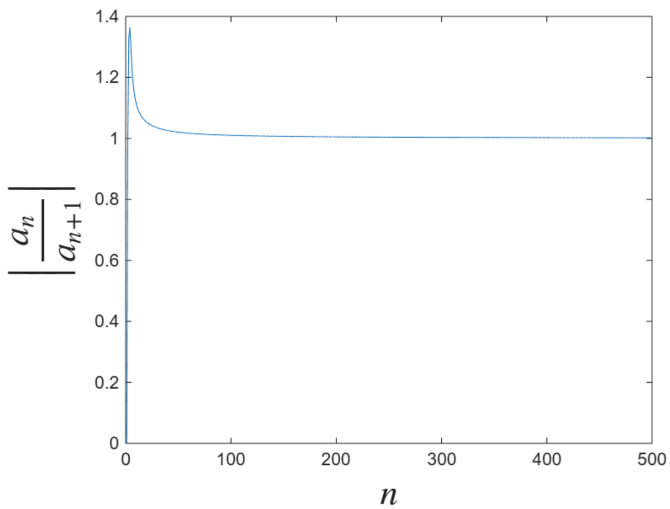

Figure 6: Ratio Test to approximate R: $\limsup |a_n/a_{n+1}|$

ii. the root test, plotting $|a_n|^{-1/n}$ versus n and seeking the limiting value as $n \to \infty$.

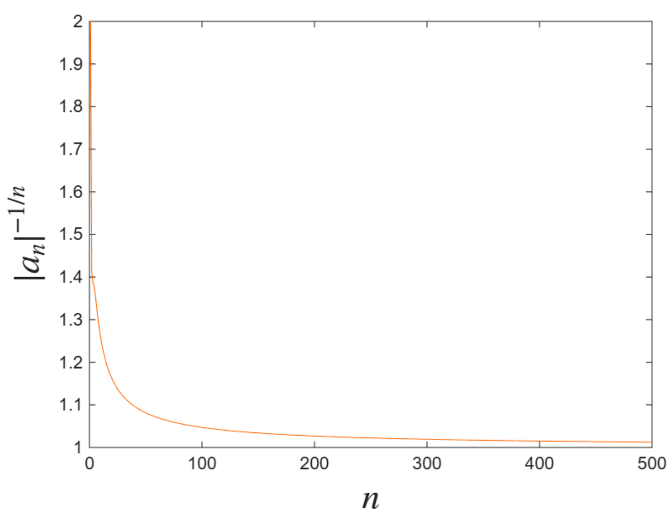

Figure 7: Root Test: $\limsup |a_n|^{-1/n}$

We see that both ratio and root test approximates $R = 1$ for large N.

iii. a Domb-Sykes plot of the ratio test, plotting $\left|\frac{a_n}{a_{n+1}}\right|$ versus $1/n$. Use the two values closest to $1/n = 0$ to linearly extrapolate to the $1/n = 0$ intercept. This intercept provides you with an estimate for R as $n \to \infty$.

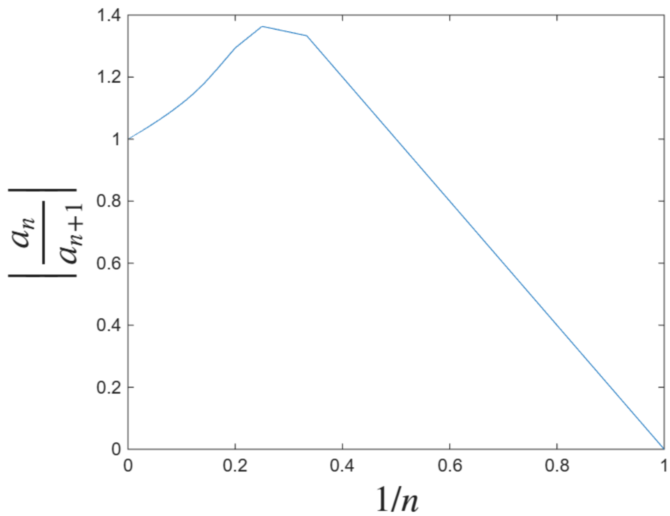

Figure 8: Domb-Sykes plot of the ratio test

The intercept of 1 in figure 8 indicates the radius of convergence is 1.

iv. a Domb-Sykes plot of the root test, plotting $|a_n|^{-1/n}$ versus $1/n$. Use the two values closest to $1/n = 0$ to linearly extrapolate to the $1/n = 0$ intercept. This intercept provides you with an estimate for R as $n \to \infty$.

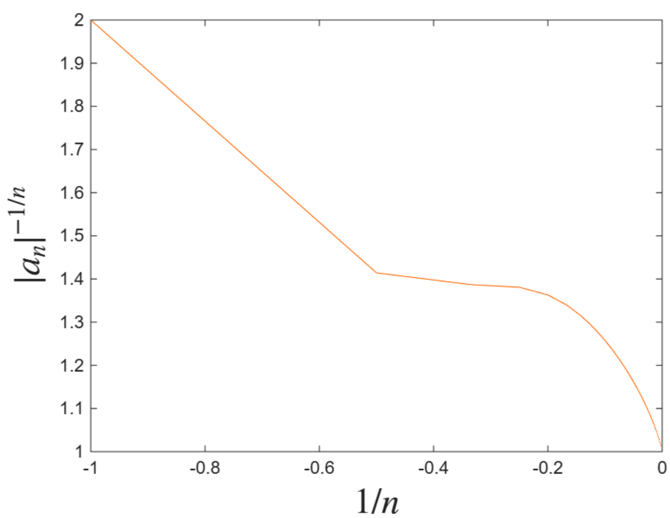

Figure 9: Domb-Sykes plot of the root test

Again at $\frac{1}{n} \approx 0$ we have the estimate for R as $n \to \infty$. Then $R \approx 1$ for both Domb-Sykes plots.

(b) In (a), how did you decide how many terms (N) to take in the series for each test?

We choose $N = 500$ via trial and error as there is clear convergence in the ratio and root test, thus surely we have convergences on the intercept for both the Domb-sykes plots. From agreements of all four plots we have radius $R \approx 1$.

MATLAB Codes

MATLAB - Problem 1

```
figure(1)
dx=0.001;
x=-3:dx:3;
f=exp(x)./(1+x);
figure(1)
plot(x,f,'LineWidth',2,'DisplayName','Actual Function'); axis1=axis; ylim([-30 30]);
hold on
xlabel('$x$','Interpreter','latex','FontSize',20)
ylabel('$f$','Interpreter','latex','FontSize',20)

N=100;
at=zeros(1,N);
at0=1; at(1)=1;
b0=1; b(2)=1;
c0=1;c1=0;
for n=1:N
    c(n+1)=1./gamma(n+2) - c(n);

axis1=axis;
fS=c0;
for n=1:N
    fS=fS+c(n)*x.^n;
    figure(1)
    if round(n/10)==n/10  %this skips every 10 terms
        plot(x,fS,'--','LineWidth',2,'DisplayName',[num2str(n) ' term series']);
    end
    axis(axis1);
end
legend('location','southeast'); axis(axis1)
figure(2)
nv=1:N; %array for n, going from 1 to N
ratio_test=abs(c(1:end-1)./c(2:end));
plot(nv(1:end),ratio_test)
xlabel('$n$','Interpreter','latex','FontSize',20)
ylabel('$\big|\frac{c_{n}}{c_{n+1}}\big|$','Interpreter','latex','FontSize',20)
figure(3)
nv=0:100; %array for n, going from 1 to N
root_test=abs(c).^(-1./nv);
plot(nv,root_test, 'Color', "#FF6600")
xlabel('$n$','Interpreter','latex','FontSize',20)
ylabel('$|c_n|^{-1/n}$','Interpreter','latex','FontSize',20)
```

MATLAB - Problem 2

```
figure(1)
dx=0.0001;
x=dx:dx:10;
f=x;
u0=[-2.392, 0];
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)+dx/2) ;
    k3=F(u+k2*dx/2,x(n)+dx/2) ; k4=F(u+k3*dx,x(n)+dx) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4);  f(n+1)=u(1);
end; hold on

plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',20,'displayname','Numerical (RK4)',
Color= "m");
hold on
legend show; xlabel('$x$','Interpreter','latex','FontSize',20)
ylabel('$f$','Interpreter','latex','FontSize',20)
function dudx = F(u,x)
f=u(1); v=u(2);
dudx=[v  f^2+f-v/x];
end

N=674;
a0 = -2.392;
a(1) = 0;
a(2) = (a0+a0^2)/4;
for n = 1:N-1
    asum = a0*a(n) + a(n)*a0 ;% j=0 and j=n terms
    for j = 1:n-1
        asum = asum + a(j)*a(n-j);
    end
    a(n+2) = (a(n) + asum)./ (n+2).^2;
end
fS=a0; axis1=axis;
for n=1:N
    fS=fS+a(n)*x.^(n);
    if round(n/74)== n/74 %this skips every 10 terms
        plot(x,fS,'--','LineWidth',2,'DisplayName',[num2str(n) ' term series']);
    end
end; axis(axis1)
legend('location','northeast','fontsize',10)
figure(2)
nvskip=2:2:N;
askip=a(2:2:end);
root_test=(abs(askip)).^(-1./nvskip);
plot(nvskip,root_test);
xlabel('$n$','Interpreter','latex','FontSize',20)
ylabel('$|a_n|^{-1/n}$','Interpreter','latex','FontSize',20)
```

MATLAB - Problem 3

```
clear;clc;clf
N=500;
a0=1;
a(1)=0;
for n=1:N
    a(n+1)=((-1)^n - a(n) )/ (n+1);
end
fS=a0;
figure(1)
nv=1:N; %array for n, going from 1 to N
ratio_test=abs(a(1:end-1)./a(2:end));
plot(nv(1:end),ratio_test)
xlabel('$n$','Interpreter','latex','FontSize',20)
ylabel('$\big|\frac{a_{n}}{a_{n+1}}\big|$','Interpreter','latex','FontSize',20)

figure(2)
nv=0:N; %array for n, going from 1 to N
root_test=abs(a).^(-1./nv);
plot(nv,root_test, 'Color', "#FF6600")
xlabel('$n$','Interpreter','latex','FontSize',20)
ylabel('$|a_n|^{-1/n}$','Interpreter','latex','FontSize',20)

figure(3)
nv=1:N; %array for n, going from 1 to N
ratio_test=abs(a(1:end-1)./a(2:end));
plot(1./nv(1:end),ratio_test)
xlabel('$1/n$','Interpreter','latex','FontSize',20)
ylabel('$\big|\frac{a_{n}}{a_{n+1}}\big|$','Interpreter','latex','FontSize',20)

figure(4)
nv=0:N; %array for n, going from 1 to N
root_test=abs(a).^(-1./nv);
plot(-1./nv,root_test, 'Color', "#FF6600")
xlabel('$1/n$','Interpreter','latex','FontSize',20)
ylabel('$|a_n|^{-1/n}$','Interpreter','latex','FontSize',20)
```

References

[1] G. R. Flierl, Baroclinic solitary waves with radial symmetry, Dyn. Atmos. Oceans, 3 (1979), pp. 15-38.
[2] Barlow, N. S., Stanton, C. R., Hill, N., Weinstein, S. J., & Cio, A. G. (2017). On the summation of divergent, truncated, and underspecified power series via asymptotic approximants. The Quarterly Journal of Mechanics and Applied Mathematics, 70(1), 21–48